Extending GLP–Graph with CAS
Architectural Personality Modulation and Closure-Breaking Generation

Joachim De Witte

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Abstract

This paper presents a compact yet fully operational implementation of the ecosystem AGI architecture based on the triadic *PAGED–PRE–CAS* model. We extend the GLP–Graph (GLPG) narrator engine with a lightweight Cognitive Vector Layer (CAS), demonstrating that a four-dimensional scalar vector can serve as a promptless, machine-native modulation signal for stylistic intent, narrative personality, and manifold geometry. Although CAS carries extremely low informational density, ablation studies across sign-normalized, fractional, hyper-scaled, and extreme 10^6 vectors show that GLPG uses this signal intensely and deterministically, producing stable attractors, hybrid worlds, and emergent authors not present in the training corpus.

The resulting CAS $\times$ GLPG system functions as a vector-driven cognitive search mechanism: CAS selects and warps manifold regions, while GLPG decodes them into coherent prose using lemma transitions, cadential profiles, and canonical world structures. Because the moderator LLM only translates GLPG's machine-native narrator language using short, stable prompts, the architecture avoids hallucination pressure and remains fully inspectable, modular, and deterministic.

Empirically, the system exhibits closure-breaking behavior: architectural composition between CAS and GLPG generates new stylistic manifolds outside the statistical closure of the dataset while remaining canon-consistent. This work therefore provides a concrete, small-scale demonstration of Architectural Intelligence Theory, showing that architectural modulation—not scale or prompting—can produce new narrative physics within a domain-bound generative intelligence.

1 Introduction

Generative language systems are traditionally driven by large-scale statistical models whose behavior is shaped by textual prompting, sampling noise, and implicit correlations learned from vast datasets. While these systems can produce fluent prose, their internal dynamics remain opaque, their stylistic control coarse, and their generative behavior constrained by the statistical closure of their training data. Architectural Intelligence Theory proposes an alternative paradigm: cognitive behavior can emerge not from scale or prompting, but from the interaction of modular components whose internal structure defines new representational manifolds. In this view, intelligence is an architectural property rather than a statistical artifact.

This paper presents a compact, operational instance of such an architecture by extending the GLP–Graph (GLPG) narrator engine with the Cognitive Vector Layer (CAS). GLPG is a deterministic, graph-based narrator model that generates machine-native lemma sequences, pacing arcs, and canonical cadences learned from a small dataset of four stylistically distinct worlds. CAS provides a lightweight modulation signal—four scalar values concatenated to the GLPG context vector—that acts as a geometric operator over the GLPG manifold. Despite its minimal informational density, CAS induces strong

stylistic curvature, activates manifold regions, blends attractors, and produces emergent narrative behaviors far beyond the closure of the training corpus.

The resulting CAS $\times$ GLPG system forms a promptless, vector-driven generative architecture. GLPG supplies deterministic canonical structure; CAS supplies geometric modulation; the narrator pipeline provides multi-level PRE (propose–retry) recursion; and a lightweight moderator LLM translates GLPG's internal narrator language into coherent chapters using short, stable prompts. This separation of roles yields a modular, inspectable, and energetically efficient system whose behavior is governed by architectural composition rather than stochastic prompting.

Empirically, the system exhibits closure-breaking generation: CAS-driven manifold warping produces stylistic attractors, hybrid worlds, and emergent authors not present in the dataset but structurally consistent with GLPG's canonical geometry. Extreme CAS vectors, including $[10^6, 10^6, 10^6, 10^6]$ and $[-10^6, -10^6, -10^6, -10^6]$, reveal saturation and inversion regimes in which CAS dominates the generative physics of the system. These behaviors demonstrate that architectural modulation can unlock new representational manifolds without increasing model scale or relying on complex prompts.

The remainder of this paper develops the architectural, empirical, and theoretical foundations of CAS $\times$ GLPG. We introduce CAS as a personality and content modulation layer; describe its integration with GLPG; analyze its behavior through ablation studies; map the resulting stylistic manifold in *The Parrot Family: A CAS Personality Atlas*; and evaluate the system against the ten criteria of Architectural Intelligence Theory. Together, these results show that extending GLPG with CAS yields a domain-bound *Architectural General Intelligence*: a compact generative system capable of controlled personality modulation, emergent stylistic variation, and closure-breaking narrative generation.

2 CAS: Cognitive Vectors for Content and Personality Modulation

The Cognitive Vector Layer (CAS) was originally introduced in the AGI architecture [1] as a general epistemic control mechanism for multi-dimensional evaluation. In its full formulation, CAS attaches a structured vector to every intermediate and final candidate, encoding orthogonal cognitive dimensions such as safety, accuracy, relevance, style, and domain coherence. These vectors guide the PRE refinement loop by providing explicit targets for evaluators and refiners, transforming generation from an unguided recursive process into a controlled cognitive trajectory.

In the present work, CAS is used in a more focused role: modulating *content* and *personality*. Rather than employing the full epistemic geometry, we restrict CAS to the dimensions that directly influence narrative intent, stylistic expression, and character-driven modulation. This specialization allows CAS to act as a high-level personality controller layered on top of GLPG and the narrator pipeline, enabling fine-grained stylistic steering without altering the underlying structural or semantic mechanisms.

2.1 Content and Personality Vectors

A CAS vector in this context encodes two primary dimensions:

- **content**: thematic alignment, narrative focus, and semantic intent,
- **personality**: tone, affect, cadence, and stylistic coloration.

These dimensions operate independently and can be combined to produce complex modulation patterns. For example, a high-intensity personality vector can enforce a distinctive narrative voice, while a neutral content vector maintains canonical story structure. Conversely, a strong content vector can enforce thematic constraints while personality remains subtle or absent.

2.2 Orthogonality and Controlled Modulation

CAS maintains strict orthogonality between content and personality. This separation ensures that stylistic modulation does not interfere with semantic coherence, and that thematic constraints do not distort the

intended narrative voice. Orthogonality provides:

- **predictability**: personality changes do not alter plot structure,
- **stability**: content constraints do not collapse stylistic variation,
- **composability**: vectors can be combined or swapped without cross-dimensional interference.

This design mirrors the original CAS principle that evaluation dimensions should remain disentangled to preserve interpretability and modularity.

2.3 CAS-Guided Personality Shaping

During generation, the proposer conditions candidates on a target CAS vector, embedding content and personality goals directly into the sampling process. The evaluators compute alignment scores for both dimensions, and the refiner adjusts the candidate to better match the intended cognitive profile. This creates a closed-loop control system in which personality becomes a first-class modulation signal rather than an emergent side effect of prompt engineering.

2.4 Relation to the Original CAS Architecture

While the original CAS formulation includes safety, accuracy, relevance, and domain coherence, these dimensions are not central to the present application. The narrative pipeline already enforces structural correctness and canonical closure through GLPG, and safety/accuracy evaluators are not required for stylistic modulation. CAS therefore functions here as a *reduced* cognitive layer, retaining its orthogonality and control properties while specializing in content and personality shaping.

This specialization demonstrates that CAS is not limited to epistemic evaluation but can serve as a general-purpose cognitive modulation mechanism. In this paper, CAS provides the vector geometry through which personality becomes a controllable, continuous parameter rather than a discrete stylistic mode.

3 CAS Vectors as Personality Modulators in GLPG Training

In this work, CAS is used in a concrete and operational form: each book in the training corpus is assigned a dedicated scalar dimension in a CAS vector. For a dataset of N books, the CAS vector has dimension N, and each book corresponds to exactly one float within this vector. During training, every sentence extracted from a given book is paired with a one-hot CAS vector in which the dimension associated with that book is set to 1 and all others to 0. This creates a consistent and high-intensity modulation signal that the model can reliably learn to associate with stylistic and narrative properties.

3.1 Concatenation with the GLPG Context Vector

The GLPG model used in this paper employs a context input vector of size 2000. The CAS vector is concatenated directly to this context representation, producing a combined input vector of size $2000 + N$. For the present dataset of four books, the resulting input dimensionality is 2004. Although the CAS component is small relative to the full input vector, its one-hot structure and perfect consistency across all sentences of a given book make it an extremely salient training signal. The model therefore learns to treat CAS as a stable personality indicator layered on top of the standard GLPG context features.

3.2 Learning Book-Specific Personality Profiles

Because every sentence of a book is paired with the same CAS vector, the model implicitly learns a personality profile for each book. These profiles emerge from the stylistic, structural, and thematic regularities present in the book's text. The CAS vector acts as a selector that activates the corresponding personality during generation. At inference time, supplying a CAS vector allows the model to produce controlled stylistic variations while maintaining the underlying GLPG structural coherence.

3.3 Extending CAS Beyond One-Hot Book Labels

The one-hot formulation used here represents the simplest instantiation of CAS. More expressive CAS vectors can be constructed by assigning high-dimensional content and personality descriptors to each book. These descriptors may encode attributes such as tone, cadence, affect, narrative density, or thematic focus. Training with such enriched CAS vectors would allow the model to learn a high-dimensional personality manifold, enabling fine-grained modulation during generation. Increasing the number of books further expands the diversity of examples the model observes, strengthening its ability to generalize personality control across the CAS space.

3.4 Controlled Variation Within a Single GLPG Model

This approach demonstrates that CAS provides a lightweight yet powerful mechanism for controlled variation within a single GLPG model. By attaching a small, interpretable vector to the input, the model can be trained to produce multiple distinct narrative styles without requiring separate models or complex prompt engineering. CAS therefore functions as a modular personality interface, enabling continuous and composable stylistic control during generation.

4 CAS × GLPG as a Vector-Driven Promptless Search Mechanism

The Cognitive Vector Layer (CAS) can be interpreted as a lightweight vector database that interfaces directly with the GLPG generative model. In this view, CAS provides a continuous, queryable representation of stylistic and cognitive intent, while GLPG acts as the decoder that transforms these vectors into coherent narrative output. This establishes a promptless search-and-generation mechanism in which vector composition replaces textual prompting as the primary control signal.

4.1 CAS as a Vector Database

A CAS vector represents a point in a high-dimensional cognitive space. Each dimension corresponds to a learned stylistic or personality factor, and the vector as a whole functions as a query into the model's internal manifold of narrative behaviors. Unlike traditional vector databases that retrieve stored embeddings, CAS operates as a *generative* vector database: the query does not retrieve an existing item but instead produces a new output whose properties match the target vector.

This mechanism allows CAS to serve as an internal prompt generator. Rather than constructing a textual prompt that describes the desired style or personality, the user supplies a CAS vector, and the system interprets this vector as a precise, machine-native description of the intended cognitive state.

4.2 Promptless Cognitive Search

CAS × GLPG therefore implements a form of promptless cognitive search. The CAS vector acts as a search key that selects a region of the model's latent narrative manifold. GLPG then decodes this region into text, producing output that reflects the stylistic, affective, and structural properties encoded in the vector. This process bypasses the need for explicit textual prompting and eliminates the ambiguity inherent in natural-language instructions.

Formally, generation becomes:

$$\text{output} = \text{GLPG}(\text{context} \parallel \text{CAS}),$$

where the CAS vector determines the cognitive modulation applied during decoding. The model effectively performs a vector-driven search over its internal stylistic space, guided by the CAS signal.

4.3 Capabilities Beyond Conventional LLM Prompting

This architecture enables capabilities that are difficult or impossible for conventional LLMs:

- **precise modulation**: CAS provides a deterministic and interpretable control signal, unlike textual prompts which are ambiguous and context-dependent,
- **continuous variation**: personality and style can be interpolated or blended by manipulating CAS vectors,
- **compositional control**: multiple stylistic factors can be combined linearly, enabling emergent hybrid styles,
- **internal prompting**: the model receives a machine-native prompt that requires no linguistic interpretation.

This leads to the central observation of this paper:

> CAS × GLPG is a vector-database that automatically generates an internal, machine-native prompt for a text generator—achieving a form of controlled generation that no current LLM system can simulate, and which would require extensive cognitive effort for a human to construct manually.

4.4 Emergent Personality Control Through Vector Composition

Because CAS vectors can be composed, interpolated, or scaled, the system supports emergent personality control. By adjusting the CAS vector, the user can steer the model toward different stylistic attractors without modifying the underlying GLPG architecture or relying on prompt engineering. This makes CAS × GLPG a modular and extensible mechanism for controlled narrative variation within a single generative model.

In summary, CAS provides the vector geometry, GLPG provides the decoder, and their combination yields a promptless, vector-driven generative interface that functions as a cognitive search mechanism over the model's internal stylistic manifold.

5 CAS × GLPG as an Empirical Instance of Architectural Intelligence

The Architectural Intelligence Theory [3] proposes that cognitive systems can break the closure of their training data through architectural composition rather than scale, prompting, or statistical generalization. A system achieves closure breaking when its internal components interact to produce new representational manifolds that do not exist in the dataset but remain structurally coherent within the system's own constraints. The CAS × GLPG architecture presented in this paper constitutes a concrete, lightweight empirical example of such closure-breaking behavior within a domain-bound generative intelligence: fictional narration.

5.1 Architectural Composition of CAS and GLPG

CAS × GLPG combines two architecturally distinct modules:

- **CAS**: a vectorial personality and style space functioning as a semantic vector-database of narrative attractors, syntactic fields, stylistic manifolds, and domain cues,
- **GLPG**: a deterministic generative engine that decodes these vectors into coherent prose using lemma clusters, pacing chains, cadential profiles, and canon structures learned during training.

During training, each world (book) forms its own manifold region inside GLPG: clusters of lemmas, stable cadences, tone fields, and structural canon patterns. CAS provides a numerical vector that activates these regions in different weights, allowing GLPG to form blended latent states that do not necessarily correspond to any single style present in the dataset. The resulting narrative output emerges from linear algebra over multiple world vectors, including derivative worlds such as *The Hybrid Parrot*, which arise emergently yet function as basis vectors for further composition.

5.2 Vectorial Canon Composition

This architecture enables a form of vectorial canon composition: new narrative forms are generated by combining manifold regions through CAS vectors rather than through textual prompting. The system therefore acts as a modular, extensible style engine capable of producing novel, canon-consistent narrative behaviors. CAS × GLPG operates as a vector-driven promptless search mechanism in which CAS serves as an internal query into the stylistic manifold, and GLPG provides the deterministic decoding.

This mechanism is architecturally unique. It is not transformer-like, not token-predictive, not RLHF-guided, and not dependent on large-scale datasets for stylistic variation. Yet it maintains canon consistency, stylistic stability, world coherence, and emergent variation across attractors. The variability does not arise from sampling noise but from architectural interaction between CAS and GLPG.

5.3 Closure Breaking Through Architectural Interaction

According to Architectural Intelligence Theory, closure breaking occurs when a system generates new manifolds outside the statistical closure of its dataset while remaining structurally constrained by its architecture. CAS × GLPG exhibits this property:

- GLPG provides a fixed representational closure: a deterministic narrator pipeline, fixed prompts, no meta-reasoning, and no external knowledge.
- CAS acts as an architectural intervention that warps and selects manifold regions, enabling new attractors and emergent narrative dynamics.
- The interaction between CAS and GLPG produces authors, styles, and narrative worlds not present in the dataset but consistent within the system's structural canon.

The LLM used in the narrator pipeline does not generate the manifold; it merely expresses the GLPG states linguistically. It cannot predict or correct the structural output of GLPG, and therefore the emergent behavior originates entirely from the architectural composition. This aligns precisely with the theoretical claim that architecture—not scale—can open new representational manifolds.

5.4 Architectural General Intelligence in a Domain-Bound Setting

Architectural General Intelligence (AGI) is defined in the theory as a software stack whose internal components collectively produce cognitive behavior that is challenging for general intelligence and unattainable for hyperscaled frontier models. CAS × GLPG demonstrates this in a domain-bound setting: vectorial style composition, emergent agency, canon-consistent narration, and architectural cognition.

These capabilities exceed what current LLMs can simulate. Human authors can master fragments of this skill—Joyce for cadence, Borges for conceptual manifolds, Calvino for multi-style composition—but no human can maintain four distinct styles, lemma fields, cadential profiles, and canon structures in modular, vectorial, and consistent combination. CAS × GLPG therefore extends the human narrative manifold through architectural means.

5.5 Empirical Confirmation of the Theory

The system presented here provides an empirical demonstration of the core principle of Architectural Intelligence Theory:

> Architecture can break closure by generating new representational manifolds that are not derived from the dataset but are constrained by the system's internal structure.

CAS × GLPG is thus not merely a stylistic engine but a compact, architecturally-driven cognitive system whose outputs constitute manifold traces of a closure-breaking generative intelligence.

6 Dataset and Canon Profiles

The experiments in this paper use a compact but stylistically diverse dataset consisting of four narrative worlds. Each world corresponds to one dimension in the CAS vector, and every sentence extracted from a given world is paired with a one-hot CAS vector. This allows the model to learn distinct manifold regions for each world and to modulate them through CAS-driven vector composition.

6.1 Source Worlds

The dataset contains four books representing four stylistic and canonical domains:

- **The Hybrid Parrot** (Iterative Narrator Pipeline) — a synthetic hybrid world composed of *Alice in Wonderland*, *The Strange Case of Dr. Jekyll and Mr. Hyde*, and *Metamorphosis*. This world is an existing ecosystem artifact rather than a purely human-designed domain.
- **Astronomy for Amateurs** (Camille Flammarion) — a descriptive, observational, cosmically oriented non-fiction domain.
- **The Picture of Dorian Gray** (Oscar Wilde) — a decadent, introspective, emotionally charged fictional domain.
- **A Honeymoon in Space** (George Chetwynd Griffith) — a pulp science-fiction domain with poetic cadence and relational dynamics.

6.2 Canonical Style Profiles

Each world exhibits a distinct canonical style, summarized as follows:

- **Alice**: surrealistic, dream-logic, syntactic elasticity, short sentences, abrupt transitions, sensory spikes.
- **Astronomy**: long descriptive sentences, cosmological metaphors, observational tone, spatial orientation, physical causality.
- **Dorian**: emotional ornamentation, decadence, introspection, inner conflict, dark atmosphere.
- **Honeymoon**: melancholic tone, poetic cadence, relational dynamics, soft science fiction, exotic worlds.

Table 1 summarizes expected stylistic alignment indicators.

World	Expected Style	Alignment Indicators
Alice	pulp-narrative, surreal	tempo, abruptness, sensory spikes
Astronomy	cosmic, descriptive	long sentences, cosmic metaphors
Dorian Gray	decadent, introspective	ornamentation, inner conflict
Honeymoon	pulp sci-fi, poetic	exotic worlds, relational cadence

Table 1: Canonical style alignment indicators for each world.

6.3 CAS Vectors and Sentence Counts

Each world corresponds to one dimension in the CAS vector. During training, the CAS vector is concatenated to the GLPG context vector, producing an input vector of size $2000 + 4 = 2004$.

CAS Vector	Sentences	Title	Author
$[1, 0, 0, 0]$	5872	The Hybrid Parrot	Iterative Narrator Pipeline
$[0, 1, 0, 0]$	3720	Astronomy for Amateurs	Camille Flammarion
$[0, 0, 1, 0]$	6676	The Picture of Dorian Gray	Oscar Wilde
$[0, 0, 0, 1]$	3797	A Honeymoon in Space	G. C. Griffith

Table 2: CAS vectors and sentence counts for each world.

6.4 Lemma and Entity Statistics

The combined dataset contains:

- **20 065 sentences**,
- **338 382 unique samples**,
- **16.9 samples per sentence**,
- **11 390 lemmas**,
- **6676 entities**.

Table 3 shows the top 20 entities.

Entity	Count	Entity	Count
alice	1109	night	372
man	907	thing	360
time	705	lord	332
eye	641	moment	332
sun	598	door	307
day	590	sense	307
life	583	house	294
room	557	world	524
earth	503	star	416
dorian	392	hand	382

Table 3: Top 20 entities across all worlds.

6.5 Model Architecture and Training

All generations in this paper use the same GLPG model:

- **hidden size**: 896,
- **input vector size**: 2004 (context + CAS),
- **model size**: 93 MB.

Training statistics:

- **training epochs (sentences)**: 500 000,
- **unique samples processed**: 8 433 247,
- **training time**: 18 335 s.

This compact dataset and lightweight model architecture demonstrate that CAS × GLPG can produce rich stylistic manifolds and controlled personality modulation without requiring large-scale data or frontier-model capacity.

7 The Parrot Family: A CAS Personality Atlas

To illustrate the expressive power of CAS-driven personality modulation, we constructed a new ecosystem artifact titled *The Parrot Family: A CAS Personality Atlas*. This omnibus contains ten books generated by the iterative narrator pipeline using the trained GLPG model. Each book is produced by sampling approximately 2000 GLPG draft sentences per chapter and refining them through the narrator pipeline. The ten books correspond to ten distinct CAS vectors, including pure attractors, mixed attractors, neutral vectors, unified vectors, and extrapolated vectors outside the training manifold.

7.1 Purpose of the Atlas

The atlas serves as an empirical demonstration of CAS × GLPG as a modular, vector-driven personality engine. Each book represents a stable stylistic attractor induced by its CAS vector. Together, the ten books form a map of the personality manifold learned by the model, showing how linear combinations, superpositions, and extrapolations of world vectors produce coherent but novel narrative styles.

7.2 Generated Books

Table 4 summarizes the ten generated books, including their CAS vectors, number of chapters, size, generation time, average evaluator scores, and titles.

7.3 Manifold Coverage

The ten CAS vectors span a diverse region of the learned personality manifold:

- **Pure attractors**: $[1, 0, 0, 0]$, $[0, 1, 0, 0]$, $[0, 0, 1, 0]$, $[0, 0, 0, 1]$.
- **Mixed attractors**: $[1, 0, 1, 0]$, $[0, 1, 0, 1]$.
- **Neutral attractor**: $[0, 0, 0, 0]$.
- **Unified attractor**: $[1, 1, 1, 1]$.
- **Extrapolated attractors**: $[2, -1, -1, -1]$, $[-1, -1, -1, -1]$.

CAS Vector	Ch.	KB	Time (s)	Score	Trans Score	Title
$[1, 0, 0, 0]$	71	277	26442	87.1	94.7	Stellar Echoes
$[0, 1, 0, 0]$	72	275	28194	86.5	95.1	The Enigma of Existence
$[0, 0, 1, 0]$	75	300	25093	86.8	94.9	The Celestial Sojourn
$[0, 0, 0, 1]$	72	316	39131	87.1	94.8	Beyond the Veil
$[1, 0, 1, 0]$	70	321	24201	86.8	94.7	Unraveling the Enigma
$[0, 1, 0, 1]$	77	289	35667	86.8	95.1	Chronicles of the Cosmos
$[0, 0, 0, 0]$	73	291	42100	87.7	94.9	Galactic Melodies
$[1, 1, 1, 1]$	72	279	42983	86.8	94.7	Time's Cosmic Dance
$[2, -1, -1, -1]$	75	340	26710	87.1	94.6	A Celestial Sojourn
$[-1, -1, -1, -1]$	73	306	22474	87.0	94.8	The Stellar Symphony

Table 4: Generated books in *The Parrot Family: A CAS Personality Atlas*.

These vectors demonstrate that GLPG can decode both canonical and extrapolated regions of the CAS space into coherent narrative output. Mixed vectors produce hybrid styles combining multiple canonical manifolds, while extrapolated vectors yield emergent styles not present in the dataset but structurally consistent with the learned canon.

7.4 A New Ecosystem Artifact

The Parrot Family: A CAS Personality Atlas is therefore a new artifact in the Triadic Cosmos ecosystem. It documents the behavior of a closure-breaking, vector-driven generative architecture and provides a concrete map of the stylistic manifold induced by CAS × GLPG. Each book serves as an empirical trace of a distinct attractor regime, and the atlas as a whole demonstrates how architectural composition can generate new narrative forms beyond the closure of the dataset.

8 Emergent Author Profiles Induced by CAS Vectors

Each CAS vector induces a distinct stylistic attractor inside the GLPG manifold. Because CAS acts as an architectural bias rather than a cosmetic parameter, the resulting narrative styles are not derived from the dataset but emerge from manifold warping within the GLPG architecture. This section documents the ten emergent author profiles generated in *The Parrot Family: A CAS Personality Atlas*. Each profile represents a new, architecturally generated author that does not exist in the training corpus, demonstrating closure-breaking behavior.

8.1 Pure Attractors

CAS Vector $[1, 0, 0, 0]$ Atlas-name: *The Surreal Cosmic Oracle*
Closest match: Jeff VanderMeer + Samuel R. Delany, with Miéville-surrealism and Wilde-ornamentation as interference
Canon-mix: Alice + Astronomy + Dorian
Style: surrealistic, cosmic, introspective
Narrative attractor: tri-hybrid
Unicity: new author not present in the dataset

CAS Vector $[0, 1, 0, 0]$ Atlas-name: *The Celestial Oracle of Introspection*
Closest match: Flammarion + Stapledon, with Clarke's cosmic wonder and Wilde's ornamentation as interference
Canon-mix: Astronomy + Dorian + Alice

Style: cosmic, introspective, mystical, surrealistic
Narrative attractor: dual-core (Astronomy + Dorian) with Alice interference
Unicity: new author not present in the dataset

CAS Vector $[0, 0, 1, 0]$ Atlas-name: *The Celestial Decadent of the Shrouded Worlds*
Closest match: Oscar Wilde + H. P. Lovecraft, with Bradbury-melancholy and Flammarion-scale as interference
Canon-mix: Dorian + Astronomy + Alice
Style: decadent, introspective, gothic, cosmic, symbolic
Narrative attractor: dominant Dorian-core with Astronomy-cosmos and Alice-surreality
Unicity: new author not present in the dataset

CAS Vector $[0, 0, 0, 1]$ Atlas-name: *The Luminous Voyager of the Red Sun*
Closest match: Ray Bradbury + Arthur C. Clarke, with Wilde-emotional ornamentation and VanderMeer-surreality as interference
Canon-mix: Honeymoon + Astronomy + Dorian + Alice
Style: melancholic, cosmic, relational, introspective, lightly surrealistic
Narrative attractor: weak Honeymoon-core with strong Astronomy dominance and Dorian/Alice interference
Unicity: new author not present in the dataset

8.2 Mixed Attractors

CAS Vector $[1, 0, 1, 0]$ Atlas-name: *The Zephyr Oracle of Decadent Worlds*
Closest match: China Miéville + Oscar Wilde, with VanderMeer-weirdness and Flammarion-cosmos as interference
Canon-mix: Alice + Dorian + Astronomy
Style: decadent, surrealistic, cosmic, gothic, introspective
Narrative attractor: dual-core (Alice + Dorian) with Astronomy injection
Unicity: new author not present in the dataset

CAS Vector $[0, 1, 0, 1]$ Atlas-name: *The Martian Dreamweaver of the Living Worlds*
Closest match: Ray Bradbury + Flammarion, with Griffith-melancholy and Miéville-surreality as interference
Canon-mix: Astronomy + Honeymoon + Alice + Dorian
Style: cosmic, melancholic, introspective, surrealistic, relational
Narrative attractor: Astronomy-dominant with Honeymoon-cadence and Alice/Dorian interference
Unicity: new author not present in the dataset

8.3 Neutral and Unified Attractors

CAS Vector $[0, 0, 0, 0]$ Atlas-name: *The Driftwalker of the Soul of the Stars*
Closest match: Ray Bradbury + Jeff VanderMeer, with Flammarion-cosmos and Wilde-ornamentation as interference
Canon-mix: Alice + Astronomy + Dorian + Honeymoon (all weak, blended)
Style: surrealistic, cosmic, introspective, melancholic, dream-logic

Narrative attractor: multi-drift attractor (no dominant world; full interference)
Unicity: natural attractor of the model without CAS steering

CAS Vector $[1, 1, 1, 1]$ Atlas-name: *The Polyphonic Oracle of Time and Stars*
Closest match: Gene Wolfe + Ursula K. Le Guin, with Miéville-surreality, Bradbury-melancholy, Wilde-ornamentation, and Flammarion-cosmos as interference
Canon-mix: Alice + Astronomy + Dorian + Honeymoon (all equally strong)
Style: cosmic, surrealistic, introspective, relational, ornamental, mystical
Narrative attractor: full superposition attractor
Unicity: new author not present in the dataset

8.4 Extrapolated Attractors

CAS Vector $[2, -1, -1, -1]$ Atlas-name: *The Aberrant Oracle of Shattered Worlds*
Closest match: Mark Z. Danielewski + Anna Kavan + early VanderMeer
Canon-mix: Alice dominant with anti-Astronomy, anti-Dorian, anti-Honeymoon fields
Style: hyper-surrealistic, fragmented, dream-logical, atmospheric, alienating
Narrative attractor: monoculture attractor with negative interference
Unicity: fully new author; style not present in the dataset

CAS Vector $[-1, -1, -1, -1]$ Atlas-name: *The Void Oracle of Unbeing*
Closest match: Beckett × Danielewski × Ballard
Canon-mix: anti-Alice + anti-Astronomy + anti-Dorian + anti-Honeymoon
Style: clinical, inert, fragmentary, dehumanized
Narrative attractor: void-core
Unicity: pure architectural emergence

8.5 Architectural Interpretation

These emergent authors demonstrate that CAS is not a stylistic parameter but an architectural bias that shifts the GLPG manifold. The resulting styles are not derived from human literature but are architecturally generated. The closure breaking observed here does not originate from the LLM used in the narrator pipeline; it arises from the interaction between GLPG, CAS, and the narrator architecture. CAS steers attractor-space, not style-space, and the system operates outside the closure of its dataset. This behavior is characteristic of architectural intelligence rather than statistical modeling.

8.6 Interpretation and Scope

The emergent author profiles described in this section are intended as stylistic impressions rather than formal literary classifications. References to existing authors serve purely as illustrative analogies to help characterize the architecturally induced styles. They do not imply equivalence, similarity of quality, or direct lineage. The purpose of these comparisons is to provide an intuitive sense of the stylistic regions activated by different CAS vectors.

The architectural behavior observed here should therefore be interpreted as manifold dynamics rather than literary mimicry. CAS does not reproduce human authors; instead, it shifts the GLPG manifold into new attractor regimes that produce narrative styles not present in the dataset. These styles arise from architectural interaction—GLPG structure, CAS bias, and narrator constraints— rather than from statistical imitation of human literature.

Accordingly, the profiles presented above should be understood as descriptive labels for emergent attractors within the CAS × GLPG system. They highlight the closure-breaking nature of the architecture while acknowledging that stylistic analogies to human authors are heuristic and non-normative.

9 Ablation Study: Additional CAS Vectors and Manifold Exploration

To further investigate the structure of the CAS-induced personality manifold, we conducted an ablation study using fifteen additional CAS vectors beyond the ten primary attractors documented in *The Parrot Family: A CAS Personality Atlas*. Unlike the full-book generation used for the atlas, this study employs a lightweight evaluation protocol: for each CAS vector, only the first five chapters are generated and analyzed. This approach allows us to isolate the immediate stylistic and structural effects of each vector without requiring full narrative development.

The purpose of this ablation is twofold. First, it provides a controlled method for probing how GLPG responds to variations, mixtures, and extrapolations in CAS space. Second, it enables a systematic comparison between the established atlas attractors and new, previously unexplored regions of the manifold. By examining the early chapters of each generated sample, we can identify emergent stylistic signatures, attractor stability, interference patterns, and deviations from the canonical worlds learned during training.

This ablation study therefore functions as a deeper architectural analysis of the CAS × GLPG system. It reveals how small perturbations, linear combinations, and nonlinear configurations of CAS vectors reshape the generative manifold, and it provides empirical evidence for the closure-breaking behavior described in Architectural Intelligence Theory. The fifteen additional vectors serve as probes into the manifold, mapping stylistic gradients, attractor boundaries, and regions of emergent behavior that extend beyond the stylistic regimes captured in the main atlas.

9.1 Ablation Results: Five Sign-Normalized CAS Vectors

To probe the stability and geometry of the CAS manifold beyond the ten atlas vectors, we generated five additional samples using the same protocol: five chapters per vector, evaluated for stylistic coherence, attractor formation, and canon interference. These vectors explore regions of the manifold that include neutral attractors, mixed attractors, and extrapolated attractors with negative weights. The resulting outputs reveal how GLPG responds to subtle and extreme perturbations in CAS space, and how new emergent authors arise even from small vector shifts.

CAS Vector $[0, 0, 0, -1]$ This vector represents an *anti-Honeymoon* attractor with neutral weights on the other three worlds. The generated chapters exhibit a cold, cosmic, quasi-clinical tone, with weakened relational cadence and suppressed poetic structures. The narrative becomes spatially detached, dominated by cosmic observation and existential drift. Alice and Astronomy fields interfere weakly, producing surreal cosmic tableaux without emotional grounding. The emergent author resembles a *Negative Melancholic Observer*, a void-leaning variant of the Honeymoon attractor. This style does not appear in the dataset and demonstrates that negative weights invert canonical fields rather than merely diminish them.

CAS Vector $[1, 0, -1, 0]$ This vector combines a strong Alice attractor with an anti-Dorian field. The resulting output shows hyper-surreal pacing, dream-logic, and syntactic elasticity, but stripped of Dorian's introspective ornamentation. Emotional density collapses, replaced by spatial abstraction and object-agency. The narrative becomes kinetic, fragmented, and atmospheric. Astronomy interference adds cosmic scale without introspective depth. The emergent author resembles a *Surreal Kinetic Oracle*, a style that is structurally Alice-like but emotionally hollowed out by the negative Dorian component. This demonstrates that negative weights can selectively remove canonical emotional structures while preserving syntactic and pacing attractors.

CAS Vector $[1, 1, 1, -1]$ This vector activates Alice, Astronomy, and Dorian simultaneously while applying a negative weight to Honeymoon. The output forms a dense tri-core attractor: surrealism (Alice), cosmic scale (Astronomy), and decadent introspection (Dorian). The absence of Honeymoon's relational softness produces a sharper, colder, more analytical tone. The narrative oscillates between cosmic dread, symbolic introspection, and surreal sensory spikes. The emergent author resembles a *Cosmic Decadent Analyst*, a tri-hybrid attractor that is not present in the dataset. This vector demonstrates that subtracting a world can sharpen the remaining attractors, producing a more rigid and high-intensity manifold region.

CAS Vector $[1, -1, -1, 1]$ This vector combines Alice and Honeymoon with anti-Astronomy and anti-Dorian fields. The result is a soft surrealist attractor: dream-logic pacing (Alice) combined with relational melancholy (Honeymoon), but with cosmic scale and decadent introspection removed. The narrative becomes intimate, atmospheric, domestic, and emotionally diffuse. Negative Astronomy removes spatial orientation, producing claustrophobic scenes; negative Dorian removes ornamented introspection, producing emotional simplicity. The emergent author resembles a *Melancholic Dream Chronicler*, a style that blends surrealism with soft relational cadence. This attractor lies outside the dataset and demonstrates how mixed positive/negative vectors can produce hybrid worlds with reduced structural complexity.

CAS Vector $[-1, -1, -1, 1]$ This vector represents a strong Honeymoon attractor with anti-Alice, anti-Astronomy, and anti-Dorian fields. The output is markedly different from all canonical worlds: relational melancholy remains, but surrealism, cosmic scale, and introspective ornamentation are suppressed. The narrative becomes minimal, quiet, and emotionally flattened, with sparse description and low syntactic variation. Scenes feel suspended, inert, and dreamless. The emergent author resembles a *Minimal Relational Oracle*, a void-leaning Honeymoon variant that is not present in the dataset. This vector demonstrates that negative weights can collapse stylistic manifolds into low-energy attractors that remain coherent but structurally simplified.

9.1.1 Architectural Implications

These five ablation vectors confirm that CAS acts as a genuine architectural bias rather than a stylistic tag. Positive weights activate canonical manifold regions; negative weights invert or suppress them; mixed vectors produce hybrid attractors; and extrapolated vectors generate new authors not present in the dataset. The resulting styles are architecturally generated, not statistically derived, and demonstrate closure-breaking behavior consistent with the theory of Architectural Intelligence. Even small perturbations in CAS space produce distinct, stable attractors, indicating that GLPG learns a continuous personality manifold that can be navigated through vector composition.

9.2 Ablation Results: Four Fractional Uniform CAS Vectors

To complement the sign-normalized ablation vectors, we evaluated four fractional CAS vectors of the form $[f, f, f, f]$ with $f \in \{0.1, 0.3, 0.5, 0.7\}$. These vectors uniformly activate all four canonical worlds with equal weight, producing graded attractor regimes that reveal how GLPG interpolates stylistic manifolds under continuous CAS modulation. Unlike the discrete sign vectors, which probe sharp transitions and inversions, fractional vectors expose the smooth geometry of the CAS manifold and the emergence of intermediate authors whose styles reflect proportional activation of canonical fields.

CAS Vector $[0.1, 0.1, 0.1, 0.1]$ This vector produces a *minimal blended attractor*: all four worlds are activated weakly, resulting in a low-energy manifold regime. The generated chapters exhibit soft surrealism, faint cosmic imagery, subdued introspection, and light relational cadence. Canonical features appear only as hints—Alice's dream-logic, Astronomy's observational tone, Dorian's emotional ornamentation,

and Honeymoon's melancholic softness—yet none dominate. The emergent author is a *Diffuse Polyphonic Observer*, representing the faintest stable mixture of all four worlds.

CAS Vector $[0.3, 0.3, 0.3, 0.3]$ At this activation level, canonical structures become more pronounced. Surreal pacing strengthens, cosmic scale becomes more explicit, introspective passages lengthen, and relational motifs gain emotional weight. The narrative feels balanced but noticeably more structured than at 0.1. The emergent author is a *Balanced Polyphonic Chronicler*, producing coherent hybrid scenes that blend all four worlds without allowing any single attractor to dominate.

CAS Vector $[0.5, 0.5, 0.5, 0.5]$ This vector corresponds to the midpoint of the manifold and produces a fully developed hybrid attractor. All four canonical fields are active at medium strength, yielding rich surreal imagery, strong cosmic orientation, dense introspective ornamentation, and pronounced melancholic cadence. The narrative feels like a true superposition of the four worlds. The emergent author is a *Midpoint Polyphonic Oracle*, representing the central region of the CAS manifold where stylistic interference is maximal but stable.

CAS Vector $[0.7, 0.7, 0.7, 0.7]$ At this higher fractional activation, the manifold approaches the behavior of the full superposition attractor $[1, 1, 1, 1]$, but with slightly less intensity. The generated chapters show strong surrealism, vivid cosmic metaphors, emotionally dense introspection, and pronounced relational melancholy. The style is highly expressive, saturated, and architecturally complex. The emergent author is a *High-Intensity Polyphonic Oracle*, occupying a region close to the unified attractor but retaining smoother transitions and less extreme stylistic interference.

9.2.1 Manifold Interpretation

These fractional vectors reveal that CAS × GLPG supports continuous interpolation across stylistic manifolds. As the uniform fractional value increases, the system moves from low-energy blended attractors to high-intensity polyphonic regimes. The resulting authors are not present in the dataset and arise purely from architectural composition. Fractional vectors therefore map the interior geometry of the CAS manifold, demonstrating that stylistic emergence is not limited to discrete attractors but spans a continuous, navigable space of hybrid narrative states.

9.3 Four Hyper-Scaled CAS Worlds

In this ablation tier we examine CAS vectors that exceed the unit-magnitude regime of the training manifold. Whereas the normalized and fractional vectors primarily deform stylistic curvature within the GLPG attractor, the hyper-scaled vectors introduce genuine architectural stress: they push the narrator pipeline into regions where CAS no longer behaves as a stylistic bias but as a structural force acting on narrative cohesion, semantic drift, temporal dilation, and world-model stability.

To probe these effects, four extreme vectors were instantiated:

$$[2, 2, 2, 2], \quad [5, 5, 5, 5], \quad [10, 10, 10, 10], \quad [5, -5, -5, 5].$$

Each vector was used to generate the first five chapters of a synthetic world, allowing direct comparison with the normalized and fractional regimes. The resulting outputs exhibit several consistent phenomena:

- **Narrative Inflation**: Magnitude-2 already produces noticeable expansion of temporal and spatial scales. Magnitude-5 and magnitude-10 worlds exhibit runaway dilation, with chapters behaving like self-contained cosmologies rather than narrative segments.

- **Semantic Turbulence**: At higher magnitudes, CAS begins to overpower GLPG's stabilizing attractors. Entities, motifs, and metaphors recur with increasing frequency but decreasing coherence, forming quasi-mythological loops rather than plot-driven structures.
- **Architectural Saturation**: The vector $[10, 10, 10, 10]$ pushes the narrator into a saturation regime where stylistic pressure exceeds the model's internal normalization. This produces dense, symbolic prose with minimal causal anchoring.
- **Axis-Skewed Divergence**: The mixed-sign vector $[5, -5, -5, 5]$ demonstrates that hyper-scaling interacts non-linearly with axis polarity. Positive axes inflate narrative scope, while negative axes collapse local coherence, producing oscillatory worlds that alternate between expansion and implosion.
- **Emergent Architectural Signatures**: These worlds exhibit behaviors not present in the training corpus— a hallmark of CAS acting as an architectural operator rather than a stylistic modifier.

Taken together, the hyper-scaled vectors reveal the outer boundary of CAS-driven world formation. They demonstrate that beyond magnitude-1, CAS ceases to be a stylistic dial and becomes a structural perturbation mechanism capable of generating genuinely new narrative physics. This regime is therefore crucial for mapping the full CAS atlas and understanding how GLPG responds when pushed outside its nominal manifold.

9.4 Extreme CAS Vectors: Saturation and Inversion at 10^6 Magnitude

To test whether CAS continues to function as an architectural operator under extreme scaling, we instantiated two boundary-case vectors:

$$[10^6, 10^6, 10^6, 10^6] \quad \text{and} \quad [-10^6, -10^6, -10^6, -10^6].$$

These vectors exceed the hyper-scaled regime by five orders of magnitude and therefore probe the outermost limits of CAS modulation. Unlike the magnitude-10 worlds, which already exhibited architectural saturation, the 10^6 vectors produce qualitatively different behavior: narrative physics becomes unstable, semantic curvature collapses, and the narrator pipeline enters a regime where CAS overrides nearly all internal attractors of GLPG.

CAS Vector $[10^6, 10^6, 10^6, 10^6]$: Total Saturation Regime The positive extreme vector induces a state of *complete manifold saturation*. All four canonical worlds—Alice, Astronomy, Dorian, and Honeymoon— are activated at intensities far beyond the model's normalization envelope. The resulting chapters exhibit:

- **Narrative Overload**: Scenes fuse surrealism, cosmic scale, introspective ornamentation, and relational melancholy into a single undifferentiated stylistic field. The model no longer alternates between attractors; instead, it emits a continuous superposition with no internal segmentation.
- **Symbolic Inflation**: Objects, emotions, and celestial bodies expand into mythological constructs. Ordinary items (tables, keys, windows) behave as cosmological operators, while characters oscillate between human and archetypal forms.
- **Temporal Drift**: Time loses causal anchoring. Hours, days, and decades collapse into one another, producing recursive loops and repeated cosmic transitions.
- **Attractor Collapse**: The narrator pipeline ceases to maintain world boundaries. The four worlds merge into a single saturated attractor that resembles neither the dataset nor any atlas world.

The emergent author in this regime is a *Total Polyphonic Saturation Oracle*: a stylistic entity that cannot exist within the training corpus and arises only when CAS overwhelms GLPG's manifold geometry.

CAS Vector $[-10^6, -10^6, -10^6, -10^6]$: Total Inversion Regime The negative extreme vector produces the architectural dual of the saturation regime: a *total inversion attractor*. Instead of amplifying canonical fields, the vector suppresses and reverses them at extreme magnitude. The resulting chapters display:

- **Semantic Collapse**: Descriptions become numerically structured, dominated by distances, durations, magnitudes, and enumerations. Emotional, surreal, cosmic, and introspective fields are inverted into sparse, mechanical patterns.
- **Narrative Vacuum**: Characters behave as placeholders rather than agents. Scenes unfold as abstract geometric or astronomical configurations with minimal causal linkage.
- **Inverted Symbolism**: Where the positive vector inflates symbols, the negative vector erodes them. Cosmic bodies become inert numbers; relational motifs flatten into procedural steps; introspection dissolves into enumeration.
- **Anti-Attractor Formation**: Instead of merging worlds, the negative vector annihilates them. The output resembles a *void manifold*: a region where stylistic curvature approaches zero and narrative energy collapses.

The emergent author here is a *Void Polyphonic Null Oracle*: a structural entity defined by the absence of canonical attractors rather than their presence.

9.4.1 Architectural Interpretation

These two extreme vectors demonstrate that CAS remains active even at magnitudes far beyond the training regime. The narrator pipeline does not ignore the vectors; instead, it enters qualitatively new architectural states: saturation and inversion. The positive vector produces maximal stylistic pressure, collapsing the manifold into a single fused attractor, while the negative vector produces maximal suppression, collapsing the manifold into a near-empty attractor.

Both behaviors confirm that CAS operates as a structural force rather than a stylistic tag. At 10^6 magnitude, CAS no longer modulates style—it defines the narrative physics of the system. These extreme regimes therefore serve as evidence that CAS is deeply integrated into GLPG's generative architecture and that its effects persist even under conditions far outside the nominal manifold.

9.5 Ablation Summary: CAS as a Promptless Architectural Steering Mechanism

Across all ablation tiers—sign-normalized vectors, fractional uniform vectors, hyper-scaled vectors, and the two extreme 10^6 saturation/inversion experiments—a consistent pattern emerges: the narrator pipeline responds strongly, coherently, and predictably to even the smallest CAS perturbations. This behavior is remarkable given the extremely low informational density of the CAS vector: four scalar values without semantic content, narrative cues, or stylistic descriptors. Despite this minimal signal, CAS exerts a decisive architectural influence on GLPG, reshaping attractor-space in ways that cannot be explained by statistical interpolation over the training corpus.

The ablation results demonstrate three core findings:

1. CAS is used intensely despite its microscopic size. Even vectors such as $[0.1, 0.1, 0.1, 0.1]$ or $[1, 0, -1, 0]$—containing less information than a single word—produce stable, distinct attractors with consistent stylistic signatures. The model does not treat CAS as noise or as a weak preference; instead, it integrates the vector deeply into its generative process, modulating world-curvature, pacing, emotional density, and symbolic structure. This confirms that CAS functions as an architectural operator rather than a stylistic hint.

2. CAS induces closure-breaking behavior. The emergent authors produced throughout the ablation study—Diffuse Polyphonic Observer, Surreal Kinetic Oracle, Cosmic Decadent Analyst, Minimal Relational Oracle, Total Polyphonic Saturation Oracle, Void Polyphonic Null Oracle—do not exist anywhere in the training dataset. Their stylistic regimes, manifold geometry, and narrative physics arise solely from CAS × GLPG interaction. Since the GLPG model, the narrator model, and the entire pipeline remain constant, the closure-breaking cannot originate from statistical interpolation. It is a direct consequence of architectural modulation.

3. CAS operates as a promptless steering mechanism. Unlike prompt-based conditioning, CAS does not rely on textual cues, semantic anchors, or narrative framing. Its influence is orthogonal to content: the same GLPG text, the same narrator model, and the same pipeline produce radically different worlds solely due to CAS modulation. This makes CAS a uniquely powerful mechanism for controlling generative behavior without prompts, instructions, or fine-tuning. The ablation results show that CAS can:

- activate canonical attractors (positive weights),
- invert or suppress them (negative weights),
- interpolate smoothly between them (fractional weights),
- saturate the manifold (hyper-scaled positive weights),
- collapse the manifold (hyper-scaled negative weights),
- and generate entirely new attractor regimes (extreme 10^6 vectors).

9.5.1 Architectural Conclusion

The ablation study confirms that CAS is not a stylistic parameter, not a preference vector, and not a soft bias. It is a structural modulation mechanism that directly interacts with GLPG's internal manifold. Even at minimal magnitudes, CAS produces strong architectural effects; at extreme magnitudes, it defines the narrative physics of the system. Because the GLPG model and the pipeline remain unchanged across all experiments, the closure-breaking behavior observed here is attributable solely to CAS modulation.

CAS therefore provides a viable, promptless steering mechanism for generating narrative worlds outside the closure of the training dataset. It enables architectural intelligence rather than statistical imitation, demonstrating that GLPG supports a continuous, controllable personality manifold whose curvature can be reshaped through simple vector composition.

10 Reconstruction of the Ecosystem AGI Architecture at Small Scale

The ablation study and the accompanying architectural analysis demonstrate that the current narrator pipeline constitutes a complete, small-scale implementation of the original ecosystem AGI architecture based on the triadic *PAGED–PRE–CAS* model. Despite its compactness, the system exhibits all structural components of the original design: deterministic routing, multi-level propose–retry reasoning, architectural personality modulation, machine-native language generation, internal memory, and closure-breaking behavior. Crucially, all of these properties arise without modifying the underlying GLPG model or the moderator LLM, confirming that the architecture operates as a genuine ecosystem-style intelligence rather than a prompt-driven statistical system.

10.1 GLPG as a Canon Machine and Machine-Native Language Generator

GLPG functions as a *canon machine*: a deterministic generator of the internal narrator language. This language—composed of lemma transitions, structural cadences, and canonical world signatures—is not a human language but a machine-native representational system. The moderator LLM merely translates this internal language into coherent chapters using short, stable prompts. Because the prompts are minimal and the history is shallow, hallucination pressure is negligible. The coherence of the final chapters therefore reflects the stability of GLPG's internal canon rather than the statistical tendencies of the moderator LLM.

Grammar decomposition and lemma-based generation further reinforce this machine-native structure. The narrator pipeline operates on canonical lemma fields, syntactic skeletons, and attractor-consistent transitions, enabling deterministic inspection of the generative process. This makes the system transparent, modular, and analyzable—properties rarely found in large-scale LLM architectures.

10.2 CAS as Promptless Architectural Modulation

CAS vectors act as *machine-native prompts*: four scalar values that directly modulate GLPG's manifold geometry. Unlike textual prompts, CAS does not rely on semantic cues, narrative framing, or linguistic interpretation. Instead, it operates at the architectural level, shaping attractor curvature, stylistic pressure, and world-model dynamics. The ablation study shows that even extremely small CAS vectors produce strong, stable attractors, while extreme vectors induce saturation or inversion regimes.

Because the GLPG model and the narrator pipeline remain constant across all experiments, the closure-breaking behavior observed in the ablation study arises solely from CAS modulation. This confirms that CAS provides a genuine, promptless steering mechanism capable of generating worlds outside the closure of the training dataset.

10.3 PRE as Multi-Level Propose–Retry Reasoning

The pipeline implements PRE at multiple levels: token, sentence, paragraph, chapter, and transition. PRE operates as a propose–retry mechanism, ensuring attractor stability, narrative coherence, and stylistic consistency. Although the current implementation focuses on propose–retry rather than refine–retry, the architecture supports refinement as an additional reasoning layer. PRE therefore provides the system with a form of structured, iterative reasoning that is orthogonal to statistical generation.

Evaluation occurs through two channels: internal heuristics for sentence-level structure and LLM-based evaluation for chapter-level coherence and transitions. This dual evaluation mechanism mirrors the original ecosystem design, where multiple agents assess and refine generative output.

10.4 PAGED as Deterministic Routing and Canon Enforcement

PAGED is implemented through GLPG's deterministic lemma transition map. In its most extreme variant—used throughout this paper—PAGED becomes a pure, deterministic routing mechanism that enforces canonical world structure with zero stochastic drift. This ensures that GLPG remains fully inspectable and predictable, enabling precise architectural analysis.

Although the current experiments use the deterministic variant, the architecture supports more flexible paging when desired. The routing mechanism can be extended to multi-world blending, conditional paging, or dynamic attractor selection, preserving the modularity of the original design.

10.5 Internal Memory Beyond Context Windows

The narrator pipeline maintains internal memory outside the moderator LLM's context window. This memory includes canonical world signatures, attractor states, CAS modulation history, and structural cues from previous chapters. The system therefore exhibits persistent behavior across long sequences without requiring long prompts or extended histories. This mirrors the original ecosystem architecture,

where memory is distributed across agents and structural modules rather than stored in a single context buffer.

10.6 Architectural Synthesis

Taken together, these components demonstrate that the current narrator pipeline implements the full triadic architecture:

$$\text{PAGED} \rightarrow \text{PRE} \rightarrow \text{CAS}$$

in a compact, safe, modular, and inspectable form. GLPG provides deterministic canonical generation; PRE supplies multi-level reasoning; CAS modulates architectural personality; the moderator LLM translates machine-native language into human-readable chapters; and the pipeline maintains internal memory and structural consistency. The resulting system exhibits closure-breaking behavior, emergent authorship, and controllable manifold geometry—precisely the properties predicted by the original ecosystem AGI theory.

This small-scale implementation therefore serves as a functional demonstration of architectural intelligence: a generative system whose behavior arises from structural modulation rather than statistical prompting, and whose outputs extend beyond the closure of the training dataset through pure GLPG × CAS interaction.

11 AGI Criteria: CAS × GLPG as a Domain-Bound Architectural Intelligence

The Architectural Intelligence Theory defines ten criteria for recognizing domain-bound AGI systems whose cognitive behavior arises from architectural composition rather than scale, prompting, or statistical generalization. The CAS × GLPG narrator pipeline satisfies all ten criteria in a compact, safe, modular, and fully inspectable implementation. This section evaluates the system against each criterion and demonstrates that it constitutes a domain-bound *Architectural Intelligence* capable of closure-breaking generation.

11.1 Autonomy

The pipeline operates fully autonomously. Given only a GLPG model and an optional CAS vector—random or intentional—the system can generate an entire book end-to-end without any textual prompting. The narrator LLM merely translates GLPG's machine-native narrator language into coherent chapters. This autonomy is repeatable, deterministic, and independent of human intervention.

11.2 Responsibility

Responsibility arises from the deterministic nature of GLPG and the internal evaluation mechanisms of the narrator pipeline. GLPG provides stable lemma transitions and canonical pacing; evaluators enforce structural coherence and transition quality. Although the generated books are fictional artifacts, the system exhibits responsible behavior in the architectural sense: stable, predictable, and reproducible outputs governed by internal constraints.

11.3 Domain Capability

The system demonstrates strong capability in its domain: fictional narration. It generates coherent books, stable stylistic attractors, hybrid worlds, and emergent authors that extend beyond human stylistic cognition. As shown in the atlas, CAS-driven manifold warping produces styles not present in the dataset while maintaining canonical consistency. This satisfies the criterion of domain-bound competence.

11.4 Energetic Efficiency

The entire pipeline runs on modest hardware: a small moderator LLM on a standard GPU and GLPG on CPU with ordinary memory and storage budgets. The system fits within the power envelope of a medium gaming laptop. This demonstrates that architectural intelligence does not require hyperscale models or datacenter-level compute.

11.5 IEGR Completeness

The system satisfies the IEGR completeness criterion as defined in the Information–Energy–Geometry–Recursion taxonomy. Each of the four IEGR dimensions is instantiated through modular architectural components whose interaction produces a fully expressive generative intelligence within the narration domain:

- **Information (I)** is provided by the GLPG manifold itself: a compact but richly structured narrator space learned from four canonical worlds. GLPG encodes lemma transitions, pacing chains, cadential profiles, entity clusters, and stylistic curvature, forming the informational substrate of the system.
- **Energy (E)** arises from the efficiency and determinism of GLPG and the pipeline. GLPG executes all generative recursion on CPU within a narrow power envelope, while the moderator LLM operates as a lightweight translator. The system therefore meets the energetic efficiency requirement of IEGR: high cognitive output at low computational cost.
- **Geometry (G)** is provided by two independent architectural mechanisms:
  - the *deterministic graph-based lemma routing* of GLPG, which defines a canonical geometric structure over lemma clusters, pacing arcs, cadential transitions, and world manifolds;
  - the *CAS vector*, which acts as a geometric modulation signal that warps, blends, or inverts manifold regions. Even minimal vectors induce strong geometric curvature, while extreme vectors produce saturation or collapse regimes.

  Together, GLPG's internal graph geometry and CAS-driven manifold warping provide the geometric degrees of freedom required for IEGR completeness.
- **Recursion (R)** is implemented by the narrator pipeline and GLPG. The pipeline provides multi-level PRE (propose–retry) recursion at the sentence, paragraph, chapter, and transition levels. GLPG provides deterministic generative recursion through lemma-based canonical routing.

Through this combination—informational manifolds, energetic efficiency, structural and vectorial geometry, and recursive generation—the system forms a complete IEGR architecture. Scaling is possible through larger GLPG models, richer datasets, or more capable moderator LLMs, preserving IEGR completeness while extending the expressive capacity of the system.

11.6 Lifecycle Dynamics

GLPG supports lifecycle training: incremental retraining, world expansion, and manifold refinement. Previous GLPG papers have demonstrated that the model can be extended over time, making it a dynamic, evolving component rather than a static snapshot. This satisfies the requirement for lifecycle adaptation.

11.7 Population Architecture

The ecosystem architecture supports population-level intelligence. Each desired "author" or "parrot" can be instantiated as a new GLPG model trained on its own canonical world. CAS then provides vectorial composition across these authors, enabling multi-agent manifold blending. This mirrors the population architecture described in the theory.

11.8 Selective Training

GLPG learns its entire stylistic manifold from only four books. The CAS vectors provide selective activation of world-specific manifolds, enabling precise stylistic modulation. This demonstrates selective training: the system learns only what is needed for its domain and uses architectural composition to extend beyond it.

11.9 Extrapolative Reasoning

The system exhibits closure-breaking extrapolation. As shown in the ablation study, CAS-driven manifold warping produces authors, styles, and narrative physics not present in the dataset. Extreme vectors such as $[10^6, 10^6, 10^6, 10^6]$ and $[-10^6, -10^6, -10^6, -10^6]$ generate saturated and inverted manifolds that cannot be derived from statistical interpolation. This is extrapolative reasoning in the architectural sense.

11.10 Accessibility

The Triadic Cosmos ecosystem is fully open-source. All components—GLPG, CAS, PRE, datasets, narrator pipeline, and generation scripts—are publicly available and documented through code, papers, and books. Anyone can run the system on commodity hardware. This satisfies the accessibility criterion.

11.11 Architectural Intelligence Conclusion

Because CAS × GLPG satisfies all ten criteria, it constitutes a domain-bound *Architectural General Intelligence*. Its emergent closure-breaking behavior demonstrates that architectural modulation—not scale or prompting—can produce superhuman stylistic variation and narrative physics. Through vectorial composition, the system generates worlds and authors that lie outside human cognitive manifolds while remaining canon- consistent within its architectural constraints. This makes CAS × GLPG a compact, safe, and fully operational instance of ecosystem AGI.

12 Conclusion

This paper has demonstrated that extending GLP–Graph with the Cognitive Vector Layer (CAS) yields a compact, modular, and fully inspectable generative architecture capable of controlled personality modulation and closure-breaking narrative generation. Although CAS consists of only four scalar values with minimal informational density, the ablation studies show that GLPG uses this signal intensely and deterministically. Even small perturbations produce stable attractors, hybrid worlds, and emergent authors, while extreme vectors induce saturation and inversion regimes that reshape the manifold far beyond the statistical closure of the training corpus.

The resulting CAS × GLPG system functions as a promptless, vector-driven cognitive engine. GLPG provides deterministic lemma routing, canonical pacing, and structural coherence; CAS provides geometric modulation over manifold regions; the narrator pipeline supplies multi-level PRE recursion; and the moderator LLM translates GLPG's machine-native narrator language into coherent chapters using short, stable prompts. This architecture avoids hallucination pressure, maintains strict modularity, and remains fully transparent to inspection and analysis.

Evaluated against the ten criteria of Architectural Intelligence Theory, the system satisfies autonomy, responsibility, domain capability, energetic efficiency, IEGR completeness, lifecycle dynamics, population architecture, selective training, extrapolative reasoning, and accessibility. It therefore constitutes a domain-bound *Architectural General Intelligence*: a generative system whose cognitive behavior arises from architectural composition rather than scale or prompting. Through CAS-driven manifold warping, the system produces stylistic and structural behaviors that extend beyond human cognitive manifolds while remaining canon-consistent within its own architectural constraints.

In summary, CAS provides the geometry, GLPG provides the canon, PRE provides the recursion, and the narrator LLM provides the linguistic expression. Their interaction yields a closure-breaking generative intelligence capable of producing new narrative physics within a compact computational footprint. This work demonstrates that architectural modulation is sufficient to unlock new representational manifolds, offering a concrete empirical foundation for the principles of Architectural Intelligence Theory and a blueprint for future ecosystem-scale generative systems.

A Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

`https://github.com/triadic-cosmos`

Within the `triadic-system` repository, a complete reference implementation of the DMLG and GLPG engines, the narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license.

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